

The Kari–Moutot excess constant is at most sixty-nine

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Abstract

For a finite alphabet A and a rectangle $D = [n] \times [m]$, a set of patterns $P \subseteq A^D$ defines the two-dimensional subshift of finite type $V(P)$ of all colourings of $\mathbb{Z}^2$ every $n \times m$ window of which lies in P . Kari and Moutot proved that $|P| \leq nm$ forces a periodic configuration whenever $V(P) \neq \emptyset$, and asked for the least *excess* $c \geq 0$ such that some P with $|P| \leq nm + c$ has $V(P)$ aperiodic. No constant upper bound was known: the classical simulations of Wang tiles by pattern sets pay a per-row and per-column cost linear in the window dimension.

We prove that the constant is finite and small:

$$1 \leq c \leq 69.$$

The witness is a set of 73 patterns on a 2×2 window. It is built from a seventeen-tile Wang tile set of Kari’s multiplier type, reconstructed from scratch: rows carry balanced (Beatty) representations of reals in $[1, 3]$ and multiply them by $3/2$ or $1/2$; aperiodicity is a four-line telescoping argument ending in $3^a = 2^{a+b}$, and non-emptiness is an explicit Beatty configuration – no computer search enters either proof. A determinism lemma then recodes the tiling space, without loss, as a nearest-neighbour (2×2 -window) subshift on the cell alphabet of (multiplier, left-carry) pairs, whose realized window language has exactly 73 elements. The non-emptiness, the count 73, and the aperiodicity are formalized as one theorem in Lean 4 with mathlib, with no `sorry` and no added axioms. Within this construction the bound is rigid: deleting any one of the 73 patterns empties the subshift, so $c \leq 68$ is unreachable by shrinking the witness. Complementing the upper bound, exhaustive and independently cross-checked enumerations show that a binary alphabet on a 2×2 window admits *no* aperiodic $V(P)$ at any excess, and none on a 3×2 window with $c \leq 3$; the smallest finite case that could decide $c = 1$ – binary 5×2 with $|P| = 11$ – is precisely identified and remains open.

1 Introduction

A colouring $x : \mathbb{Z}^2 \rightarrow A$ is *periodic* if $x(\cdot + t) = x$ for some $t \in \mathbb{Z}^2 \setminus \{0\}$. For $P \subseteq A^{[n] \times [m]}$ let

$$V(P) = \{ x : \mathbb{Z}^2 \rightarrow A : x|_{v+[n] \times [m]} \in P \text{ for all } v \in \mathbb{Z}^2 \},$$

a subshift of finite type (SFT). Call $V(P)$ *aperiodic* if it is non-empty and contains no periodic configuration. Low pattern count forces periodicity: Kari and Moutot [4, Cor. 6]

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proved that if $|P| \leq nm$ and $V(P) \neq \emptyset$ then $V(P)$ contains a periodic configuration – a local-complexity analogue, for SFTs, of the phenomena around Nivat’s conjecture. In the concluding section of [4] they posed the quantitative question:

Determine the least $c \geq 0$ for which there exist n, m and $P \subseteq A^{[n] \times [m]}$ with $|P| \leq nm + c$ such that $V(P)$ is aperiodic.

Their Corollary 6 gives $c \geq 1$. In the other direction no constant bound was known. The two available mechanisms live at opposite scales: the algebraic (annihilator) method that powers the lower bound loses all force the moment the complexity exceeds $|D| = nm$, while every classical simulation of an aperiodic Wang tile set by a pattern SFT pays a coding cost *linear* in the window dimensions. Between “at least $nm + 1$ ” and “ $O(n)$ above nm ” nothing was known to be sensitive to a constant excess.

This paper closes that gap from above by a constant.

Theorem 1.1 (main theorem). *The Kari–Moutot excess constant satisfies*

$$1 \leq c \leq 69.$$

Explicitly: there is an alphabet A with $|A| \leq 8$ and a set P of 73 patterns of shape 2×2 over A such that $V(P)$ is non-empty and aperiodic; and $73 = 2 \cdot 2 + 69$.

The lower end of the interval is Kari–Moutot’s Corollary 6, used here only to bound the interval; everything else is self-contained. The upper bound has three independent layers of assurance: a written proof (Sections 3–4), replayable exact certificates, and a Lean 4 formalization of the full statement – non-emptiness, the pattern count 73, and aperiodicity in one theorem (Section 8).

Three further results frame the bound.

Theorem 1.2 (structural bridge; Lemma 2.3). *For every n, m and P , the SFT $V(P)$ is topologically conjugate to the Wang shift whose tiles are the elements of P , coloured by their $(n-1) \times m$ and $n \times (m-1)$ overlaps. Hence the excess problem is exactly: among aperiodic Wang tile sets, how small can the number of tiles be relative to the area of a window on which the set is representable?*

Theorem 1.3 (rigidity of the witness; Theorem 5.1). *Deleting any single pattern from the 73 empties the subshift: $V(P \setminus \{p\}) = \emptyset$ for every $p \in P$. Thus $c \leq 68$ cannot be reached by shrinking this witness, and 69 is deletion-minimal for the construction.*

Theorem 1.4 (small windows cannot do better; Section 6). *Over the binary alphabet, no P on a 2×2 window has $V(P)$ aperiodic – at any excess (exhaustive over all 2^{16} subsets, two independent implementations); and none on a 3×2 window with $c \leq 3$. Consequently any improvement of Theorem 1.1 at these shapes requires larger alphabets, and the smallest case that could decide $c = 1$ is the binary 5×2 window with $|P| = 11$ – a finite, well-posed, currently unresolved search.*

Method

The witness is not found by search. It is a Kari-type *multiplier* tile set [3] – rebuilt from its mechanism rather than imported – whose aperiodicity proof is algebraic: rows of a tiling carry balanced representations of reals, vertical adjacency multiplies the represented value by $3/2$ or $1/2$, and a doubly periodic tiling would force $(3/2)^a (1/2)^b = 1$ with $a + b \geq 1$, i.e. $3^a = 2^{a+b}$, which is impossible. The contribution of this paper on top of the classical

mechanism is quantitative: a determinism lemma showing that the tiling space survives, *without loss*, a recoding onto the cell alphabet of (multiplier, left-carry) pairs, at which point a 2×2 window sees exactly 73 patterns. Two earlier presentations of the same system gave excesses 169 and 130; both supersessions are corrections of unforced choices, not changes of mechanism, and the paper presents only the final form. (The companion negative results of Theorem 1.4 *are* searches, with exhaustiveness arguments and independent cross-checks; they enter no positive claim.)

2 Preliminaries

Convention 2.1. $[n] = \{0, \dots, n-1\}$ and $D = [n] \times [m]$. Patterns, $V(P)$, periodicity and aperiodicity are as in Section 1. The *excess* of $P \subseteq A^D$ is $|P| - nm$. A *Wang tile* is a unit square with coloured edges; a tile set T defines the *Wang shift* $X_T \subseteq T^{\mathbb{Z}^2}$ of assignments in which horizontally adjacent tiles agree on the shared vertical edge colour and vertically adjacent tiles agree on the shared horizontal edge colour. Aperiodicity of X_T is defined as for $V(P)$.

Lemma 2.2 (period implies doubly periodic). *If $V(P)$ contains a configuration with some non-zero period vector, then it contains a doubly periodic configuration; equivalently, a valid $w \times h$ torus configuration exists for some $w, h \geq 1$. Moreover a doubly periodic configuration exists if and only if some configuration has a horizontal period $(w, 0)$, $w \geq 1$.*

Proof. Let $x \in V(P)$ with $x(\cdot + t) = x$, $t \neq 0$. Write $t = k t_0$ with t_0 primitive and pick $M \in \text{GL}(2, \mathbb{Z})$ with $M t_0 = (1, 0)$, so $M t = (k, 0)$. The sheared configuration $x'(v) = x(M^{-1}v)$ has period $(k, 0)$ and lies in the SFT V' defined by the transported finite shape MD and patterns. A configuration of V' with period $(k, 0)$ is a bi-infinite sequence of rows over the *finite* alphabet $B = A^{\mathbb{Z}/k\mathbb{Z}}$, and since MD fits in a box of height H , membership in V' is a constraint on windows of H consecutive rows: a one-dimensional SFT Y over B , non-empty because it contains x' . A non-empty one-dimensional SFT contains a periodic point (its transition graph is finite and carries a bi-infinite path, hence a cycle). That point is doubly periodic, and doubly periodic points transport back through M^{-1} . For the last claim: a rank-two period lattice with basis $v_1 = (a_1, b_1)$, $v_2 = (a_2, b_2)$ contains the non-zero horizontal vector $b_2 v_1 - b_1 v_2$. $\square$

Lemma 2.3 (patterns are Wang tiles). *For any $n, m \geq 1$ and $P \subseteq A^D$, the SFT $V(P)$ is topologically conjugate to the Wang shift X_T with $T = P$, where the right (resp. left) colour of a pattern is its last (resp. first) $n-1$ columns and the top (resp. bottom) colour its last (resp. first) $m-1$ rows; when $n = 1$ (resp. $m = 1$) all horizontal (resp. vertical) colours coincide.*

Proof. Map $x \in V(P)$ to $T(x)(v) = x|_{v+D}$; adjacent images overlap exactly in the stated blocks, so $T(x) \in X_T$. Conversely, a valid Wang tiling assigns to each v a pattern $t_v \in P$ agreeing with its neighbours on all overlaps; the overlaps force the t_v to glue to a single well-defined $x : \mathbb{Z}^2 \rightarrow A$ with $x|_{v+D} = t_v \in P$, so $x \in V(P)$. The two maps are mutually inverse and shift-commuting. $\square$

Corollary 2.4. *If $V(P)$ is aperiodic then P , viewed as above, is an aperiodic Wang tile set with exactly $|P|$ tiles. The excess problem is therefore the problem of minimizing $|T| - nm$ over aperiodic Wang tile sets T and windows $[n] \times [m]$ on which T is representable as a pattern set.*

Remark 2.5 (a conditional floor). Jeandel and Rao [2] proved that an aperiodic Wang tile set has at least 11 tiles. With Corollary 2.4 this gives $|P| \geq 11$, hence $nm + c \geq 11$, for every aperiodic $V(P)$ – in particular $c = 1$ demands $nm \geq 10$. This paper uses the imported floor only to *locate* the decisive open case in Theorem 1.4; no positive result depends on it. An unconditional, self-contained substitute at the binary 2×2 window (no excess $c \leq 12$ is attainable there) is part of Theorem 6.1.

3 The seventeen-tile multiplier system

Definition 3.1 (the tile set T_{17}). Fix the interval $I = [1, 3]$ and the two multipliers

$$M_1 = \frac{3}{2} \text{ on } [1, 2], \quad M_2 = \frac{1}{2} \text{ on } (2, 3],$$

so that $f(q) = M_i q$ maps I into I . A tile is a quintuple

$$(\text{mult}, c_L, b, t, c_R), \quad c_R = c_L + \text{mult} \cdot b - t,$$

where $b, t \in \{1, 2, 3\}$ are the bottom and top digits, c_L, c_R are the left and right carries (scaled by the denominator of the multiplier so that all carries are integers, with the finite carry ranges recorded below), the horizontal colour is the pair $(\text{mult}, \text{carry})$ – so a row cannot change multiplier – and the vertical colour is the digit. Enforcing the carry ranges $(-1, \text{mult})$ realized by balanced representations (Theorem 3.3) leaves exactly **seventeen** tiles; the explicit list is frozen in the certificate and in the Lean specification (`tiles_spec`: the carry relation, $1 \leq b \leq 3$, $-1 \leq c_L \leq 2$, multiplier parity), closed by `decide`.

The mechanism is Kari’s multiply-by-two-multipliers idea [3]; the tile set and both proofs below are reconstructed from the mechanism, not imported.

Theorem 3.2 (aperiodicity). *The Wang shift $X_{T_{17}}$ contains no configuration with a non-zero period vector.*

Proof. By Lemma 2.2 it suffices to exclude a doubly periodic tiling, say of horizontal period $w \geq 1$ and vertical period $h \geq 1$. Fix a row j ; it carries a single multiplier mult_j , since the horizontal colour contains the multiplier. Summing the carry relation $c_R = c_L + \text{mult}_j b - t$ over one horizontal period, the carries telescope cyclically, $\sum_i (c_{R,i} - c_{L,i}) = 0$, leaving

$$\text{mult}_j \cdot \sum_{i=0}^{w-1} b_i = \sum_{i=0}^{w-1} t_i, \quad \text{i.e.} \quad \text{mult}_j q_j = q_{j+1},$$

where $q_j > 0$ is the mean bottom digit of row j (positive because every digit is ≥ 1), and q_{j+1} is the mean bottom digit of the row above, since the top digits of row j are the bottom digits of row $j + 1$. Going around the vertical period,

$$q_0 = \left(\prod_{j=0}^{h-1} \text{mult}_j \right) q_0, \quad \text{so} \quad \prod_{j=0}^{h-1} \text{mult}_j = 1,$$

i.e. $(3/2)^a (1/2)^b = 1$ with $a + b = h \geq 1$. This says $3^a = 2^{a+b}$, forcing $a = 0$ and then $b = 0$, so $h = 0$ – a contradiction. $\square$

Theorem 3.3 (non-emptiness). $X_{T_{17}} \neq \emptyset$. *Explicitly, for any $q \in I$ and offset $r \in \mathbb{R}$, the balanced (Beatty) row*

$$b_i = \lceil q(i+1) + r \rceil - \lceil qi + r \rceil$$

and the corresponding row for $(\text{mult} \cdot q, \text{mult} \cdot r)$ satisfy the carry relation with

$$c_i = \text{mult} \cdot \lceil qi + r \rceil - \lceil \text{mult}(qi + r) \rceil \in (-1, \text{mult}),$$

and stacking the rows of the f -orbit of any $q_0 \in I$ tiles the plane.

Proof. The identity $c_{i+1} = c_i + \text{mult} \cdot b_i - t_i$, with t_i the balanced digit of $(\text{mult} \cdot q, \text{mult} \cdot r)$, is immediate from the definition of c_i by cancellation; the stated membership $c_i \in (-1, \text{mult})$ follows since $0 \leq \text{mult} \lceil s \rceil - \text{mult} s < \text{mult}$ and $0 \leq \lceil \text{mult} s \rceil - \text{mult} s < 1$ for all real s . Digits of a balanced representation of $q \in [1, 3]$ lie in $\{1, 2, 3\}$, and the carry ranges are exactly those of Definition 3.1. Iterating rows along the f -orbit of q_0 gives a configuration all of whose local constraints are the carry relations, hence a valid tiling. The construction is instantiated, and every window of two independent parameter choices checked, in the frozen certificate. $\square$

4 The 2×2 recoding and the main theorem

The tile set has 17 tiles, but Lemma 2.3 runs in both directions: to score an *excess* we must present the tiling space as $V(P)$ on a small window over a small alphabet, and the cost of the presentation is the size of the realized window language. The naive presentation (cells = whole tiles, or the half-cells (mult, c_L, b)) realizes 173 extendable 2×2 blocks – excess 169. Both inefficiencies are removable.

Lemma 4.1 (determinism of the cell recoding). *Define the cell of a tile to be the pair (mult, c_L) , and recode a tiling $x \in X_{T_{17}}$ as the cell configuration $\pi(x)$. Then, machine-checked over the 17 tiles:*

1. *carries match across horizontal edges: $c_R(a) = c_L(b)$ whenever a may sit immediately left of b ; hence the pair of horizontally adjacent cells $((\text{mult}, c_L), (\text{mult}, c'_L))$ of $\pi(x)$ at positions $(i, j), (i+1, j)$ determines $(\text{mult}, c_L, c_R) = (\text{mult}, c_L, c'_L)$ of the tile at (i, j) ;*
2. *the triple (mult, c_L, c_R) determines the tile, with zero ambiguous triples among the 17;*
3. *dropping either component of the cell breaks determinism (4, respectively 2, ambiguous keys).*

Consequently a 2×2 window of $\pi(x)$ recovers both tiles of its left column and checks their vertical match, so π is a conjugacy from $X_{T_{17}}$ onto the 2×2 -window SFT $V(P)$ on the cell alphabet, where P is the set of 2×2 cell blocks realized by $X_{T_{17}}$. In particular this recoding is lossless – a conjugacy, not a factor map.

Proposition 4.2 (the realized language has 73 blocks). *The set P of 2×2 cell blocks realized by the Beatty tilings of Theorem 3.3 has exactly 73 elements, and this count is independent of the parameter: saturated sweeps over 30, then 400, then 24 independent parameter choices each realize all 73 blocks and no others. The certificate `cert_excess69.json` lists the 73 patterns; the count is also carried inside the Lean development (`PD_length`).*

Theorem 4.3 (main theorem). *$V(P)$, for the 73-element pattern set P of Proposition 4.2 on a 2×2 window over the cell alphabet $\{(\text{mult}, c_L)\}$ (at most 8 letters), is non-empty and aperiodic. Hence the Kari–Moutot excess constant satisfies*

$$1 \leq c \leq 73 - 4 = 69.$$

Proof. Non-emptiness: π of any Beatty tiling lies in $V(P)$ by the definition of P . Aperiodicity: by Lemma 4.1 every element of $V(P)$ decodes to a valid tiling of T_{17} , equivariantly, so a periodic element of $V(P)$ would decode to a periodic tiling, contradicting Theorem 3.2. The lower end of the interval is [4, Cor. 6]. $\square$

Remark 4.4 (Lean formalization). The aperiodicity layer is formalized: the Lean 4 + mathlib development states and proves

```
theorem kari_tiles_aperiodic (f :  $\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \text{Tile}$ ) (hf : IsTiling f) (a h :  $\mathbb{Z}$ ) (hper :  $\forall j$ 
```

for the explicit 17-tile list, with `IsTiling` defined as membership plus the two Wang matching conditions – no `sorry`, no added axiom, the only computational input being the finite tile specification closed by `decide`. Later parts of the same file carry the pattern-set layer: the theorem `excess_le_69` states $|P| = 73$, non-emptiness (witnessed by an explicit configuration `cfgCell`), and aperiodicity as one statement. The single claim of this paper that is *not* inside Lean is the optimality discussion of Section 5, which rests on exhaustive sweeps.

5 Rigidity: the witness is deletion-minimal

Theorem 5.1 (all 73 deletions die). *For every $p \in P$, $V(P \setminus \{p\}) = \emptyset$. Hence no proper subset of P is an aperiodic witness, and $c \leq 68$ is unreachable by shrinking this construction.*

Proof. Two facts about $V(P)$, established in the source campaign with exact certificates: first, $V(P)$ is the tiling space of T_{17} (Lemma 4.1), and that space is exactly the Beatty family – the closure of the configurations of Theorem 3.3, the orbit of a single real parameter under $q \mapsto \frac{3}{2}q$ or $\frac{1}{2}q$; second (Proposition 4.2), every member of the family realizes all 73 blocks. Hence for every p , no member of the family lies in $V(P \setminus \{p\})$; since $V(P \setminus \{p\}) \subseteq V(P)$ and $V(P)$ is exactly that family, $V(P \setminus \{p\}) = \emptyset$.

Where each deletion breaks is not where one would guess: deleting p destroys the decode structure of Lemma 4.1 in only 2 of the 73 cases; in the remaining 71 the multiplier machinery survives intact and the loss is *non-emptiness* – the excluded block occurs in every configuration of the space. $\square$

Remark 5.2 (optimality over the construction class). Within the class searched, 69 is optimal three ways: over all multiplier pairs $M_1 = p/q$, $M_2 = p'/q'$ with $p, q, p', q' \leq 9$ (multiplicative independence checked by prime exponents, tightest admissible intervals); over coarser cell alphabets (dropping either cell component breaks determinism, Lemma 4.1(3), and all coarser recodings fail); and over window shapes (the realized language grows steeply with the window: 462 at 3×2 , 999 at 2×3 , 2785 at 3×3 , so 2×2 is forced). These are exhaustive sweeps with stated boundaries, not theorems about all constructions; mechanisms not of multiplier type – hierarchical constructions, or the Jeandel–Rao set itself – were not scored and might beat 69.

6 Unconditional negatives at small windows

The results of this section are exhaustive searches; they are included because they are citation-free complements to Theorem 4.3 and because they isolate the decisive open case. Soundness rests on Lemma 2.2 (sweeping horizontal periods alone suffices, with the height handled exactly by cycle detection) and on a closure argument: if $V(P_0)$ is aperiodic then the set P of patterns actually occurring is closed under taking horizontal and vertical successors and predecessors, satisfies $V(P) = V(P_0)$, and is no larger – so it suffices to

enumerate *closed* sets, branching on supersets when a set closes (a subtlety whose earlier omission was caught and fixed against ground truth: a closed set can be a proper subset of another closed set).

Theorem 6.1 (binary 2×2 : never aperiodic). *Over $|A| = 2$ there is no $P \subseteq A^{[2] \times [2]}$, of any size, with $V(P)$ aperiodic. Exhaustive over all 2^{16} subsets with no search heuristics: 33,639 closed subsets, of which 33,411 admit a torus configuration and 228 have $V(P) = \emptyset$; zero survivors. Independently reproduced by the pruned closed-set enumeration, and the fast torus oracle cross-validated against a second implementation on thousands of random cases with zero verdict mismatches. In particular, citation-free: no excess $c \leq 12$ is attainable at a binary 2×2 window.*

Theorem 6.2 (binary 3×2 , excess ≤ 3 : never aperiodic). *Over $|A| = 2$ and the 3×2 window there is no aperiodic $V(P)$ with $|P| \leq 9$ (exhaustive; 6,083 closed torus-free sets at cap 9, all refuted).*

Remark 6.3 (the decisive open case). Under the conditional floor of Remark 2.5, the first window shapes where $c = 1$ is possible at all are those with $nm \geq 10$, and the smallest is binary 5×2 with $|P| = 11$. That case is finite and well-posed; the validated enumeration reaches roughly 8×10^5 nodes/second there but was never started for want of wall-clock budget, while the neighbouring binary 4×2 case at cap 11 was pushed to 1.21×10^9 nodes (1,943 orbit representatives, zero survivors) without finishing. Deciding binary 5×2 at $|P| = 11$ decides whether $c = 1$ at the smallest window where it can occur.

7 Open problems

1. Pin the constant: every value in $\{1, \dots, 69\}$ is currently possible. The construction class of this paper is exhausted (Remark 5.2); a smaller witness needs either a different aperiodic mechanism scored by the window language, or the $|C|$ -enumeration frontier at small alphabets.
2. Decide the binary 5×2 , $|P| = 11$ case (Remark 6.3).
3. Raise the lower bound beyond 1: the annihilator method dies at complexity $nm + 1$, and the real frontier identified in the companion campaign is the balancedness hypothesis of [4, Lem. 21] – a combinatorial, not algebraic, obstruction.
4. The general transfer principle behind Lemma 2.3 gives $c \leq |L_{2 \times 2}(X)| - 4$ for *any* aperiodic Wang shift X presentable on a 2×2 window, where $L_{2 \times 2}$ is the realized block language. Score the known small aperiodic tile sets – Jeandel–Rao’s 11 tiles [2], Culik’s 13 [1] – under this functional.

8 Provenance, verification model, and dependency disclosure

Provenance

The mathematics was produced by an autonomous multi-agent AI research fleet (2026-08-13 through 2026-08-18) in two adversarially paired windows: **kari-construct** (upper bounds; the results of Sections 3–6) and **kari-obstruct** (lower bounds and refutation; the companion results quoted in Section 7). The pairing discipline forbade either window from reading the other’s code; exit conditions were committed before computations; every

number ships with a frozen checker and certificate, replayable end-to-end by one script (`REPLAY.sh`). The campaign registered 114 claims in the fleet ledger.

Verification model

Theorems 3.2, 3.3 and Lemmas 2.2, 2.3 are conventional proofs, reproduced in full above. Lemma 4.1 and Proposition 4.2 are finite exact checks over the 17 tiles and the frozen pattern list (`cert_excess69.json`); Theorem 5.1 combines them with the space classification established in the campaign notes. The Lean 4 + mathlib development (Remark 4.4) covers non-emptiness, the count 73, and aperiodicity as one theorem with no `sorry` and no added axioms; its tile list was checked to match the generating code exactly, and the aperiodicity layer compiled clean in every operator verdict from round three onward. The exhaustive negatives of Section 6 are sweeps: their soundness arguments (closure, and Lemma 2.2) are proofs, their exhaustiveness is stated with exact node and set counts, and the enumeration bug found and fixed during cross-validation is disclosed in Section 6 rather than hidden. No sweep supports any aperiodicity claim: the repository’s standing rule, kept here, is that aperiodicity is only ever asserted on the algebraic proof.

Load-bearing literature inputs

Kari–Moutot [4, Cor. 6]: only for the lower end $c \geq 1$ of the interval (their question; their bound). Jeandel–Rao [2]: only inside Remark 2.5 and Remark 6.3 (locating the decisive case); no positive or negative result of this paper depends on it. Kari [3]: the multiplier *mechanism* is his; the tile set, both proofs, and all counts here are reconstructed independently and verified as above. Theorem 4.3’s upper bound is otherwise self-contained.

References

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